University ERP Report

By Pratyaksh (2024431) and Sourabh (2024563)

For Advanced Programming Monsoon 2025

Introduction:

This is a complete and comprehensive University ERP System built with Java and keeping in mind security and robustness.

Overview:

The system consists of three views / interfaces, namely **Admin , Student and Instructor**, for each respective role. Which are all accessible by a single login interface.

The login system is enforced by a **userID** and **passwd** , the user is redirected to their respective interface via the userType stored internally in the database, leaving NO CHANCE for any user getting redirected to wrong interface. As the interface opening is a internal decision NOT directly dependent on user inputs.

The password is encrypted and stored via the **password4j library's bcrypt2 algorithm**, ensuing no plaintext passwords are stored in the database.

Tech Stack

Language: Java
UI Framework: Java Swing, FlatLaf
Database: MariaDB
Password: Password4j library for BCrypt Password Hahing

Database Design

Databases are split into two, to ensure security and boundary between credential storage and general operational data.

- ## A. Authentication Database (authDB)

→ This database handles identity management, It contains the user table, which stores:

Column Name	Data Type	Description
userID	INT AUTO_INCREMENT	Unique identifier for each user. Automatically increments.
userName	VARCHAR(100) UNIQUE	Stores unique user names. Cannot be null.
role	ENUM("ADMIN", "INSTRUCTOR", "STUDENT")	Defines the user's role within the system. Must be one of the specified roles.
passwdHash	VARCHAR(255)	Stores the hashed password using the BCrypt algorithm. Cannot be null.
status	ENUM("ACTIVE", "INACTIVE") DEFAULT "INACTIVE"	Indicates whether the user account is active or inactive. Defaults to "INACTIVE".
lastLogin	TIMESTAMP NULL	Records the timestamp of the user's last login. Can be null if the user has never logged in.

This table is foundational for managing user authentication, ensuring that each user has a unique identity and specified access rights within the ERP system.

- ## B. ERP Database

→ This databse manages university's academic records, it contains the following tables:

Table Name	Description	Attributes
settings	Stores system-level settings, with settingKey as the crucial identifier for each setting.	settingKey, settingValue
students	Contains student information. Important attributes include userID (unique student identifier), roll (unique roll number), and program (academic program).	userID, roll, firstName, lastName, program, cgpa, passOutYear
instructors	Houses data about instructors and their affiliations. Key attribute: userID (unique instructor identifier).	userID, fullName, department
courses	Lists available courses. Significant attributes are courseID (unique course identifier) and courseCode (unique course code).	courseID, courseCode, courseName, credits
sections	Defines course sections, linking courses to instructors. Important attributes include sectionID (unique section identifier) and courseID (links to courses).	sectionID, courseID, instructorID, term, room, schedule, capacity, dropDeadline
enrollments	Manages student enrollments. Key attributes are enrollmentID (unique enrollment identifier) and studentID (links to students).	enrollmentID, studentID, sectionID, status
grades	Records student grades, ensuring detailed performance tracking. Important attributes include	gradeID, enrollmentID, componentName, score, maxScore

	gradeID (unique grade identifier) and enrollmentID (links to enrollments).	

This design ensures a clear separation between different entities and their relationships, facilitating efficient data management and retrieval. Each table is linked through foreign keys to maintain data integrity across the system.

Features

- ## Role Based Access Control
  - To ensure data security, strict role enforcement is implemented at the application entry point. The authService vertiifes credentials against authDB. Upon successful authentication, the LoginScreen directs the user to a role-specific dashboard:
  - **Student Role:** Restricted to StudentDashboard. Access is limited to the student's own records (grades, schedule) and the public course catalog.
  - **Instructor Role:** Restricted to InstructorDashboard. Write access is granted **only** for sections assigned to the specific instructor ID.
  - **Admin Role:** Grants access to AdminDashboard for user creation, course scheduling, and system maintenance.
- ## Grading Algorithm Implementation
  - The system automates the calculation of final grades using a specific weighted average logic encapsulated in InstructorService.java.
  - **The Weighting Rule:** The final "Overall" grade is computed as the sum of three distinct components:
  - **Quiz:** Maximum 20 marks.
  - **Midterm:** Maximum 30 marks.
  - **Final Exam:** Maximum 50 marks.
  - **Algorithmic Logic:** When an instructor triggers the computation, the service layer iterates through the grades table for a specific enrollment, sums the scores of the components mentioned above, and persists the result as the "Overall" grade.
- ## Maintenance Mode Enforcement
  - **Mechanism:** A boolean flag (maintenanceOn) is stored in the erpDB.settings table.
  - **Enforcement:** Critical "write" operations in the Service Layer are protected by guard clauses.
    - StudentService: Prevents registerStudent() and dropSection() execution if the flag is true.
    - InstructorService: Prevents saveOrUpdateGrade() execution if the flag is true.
  - **UI Feedback:** Dashboards actively poll this setting and display a high-contrast red banner to alert users that write actions are temporarily disabled.

Screenshots:

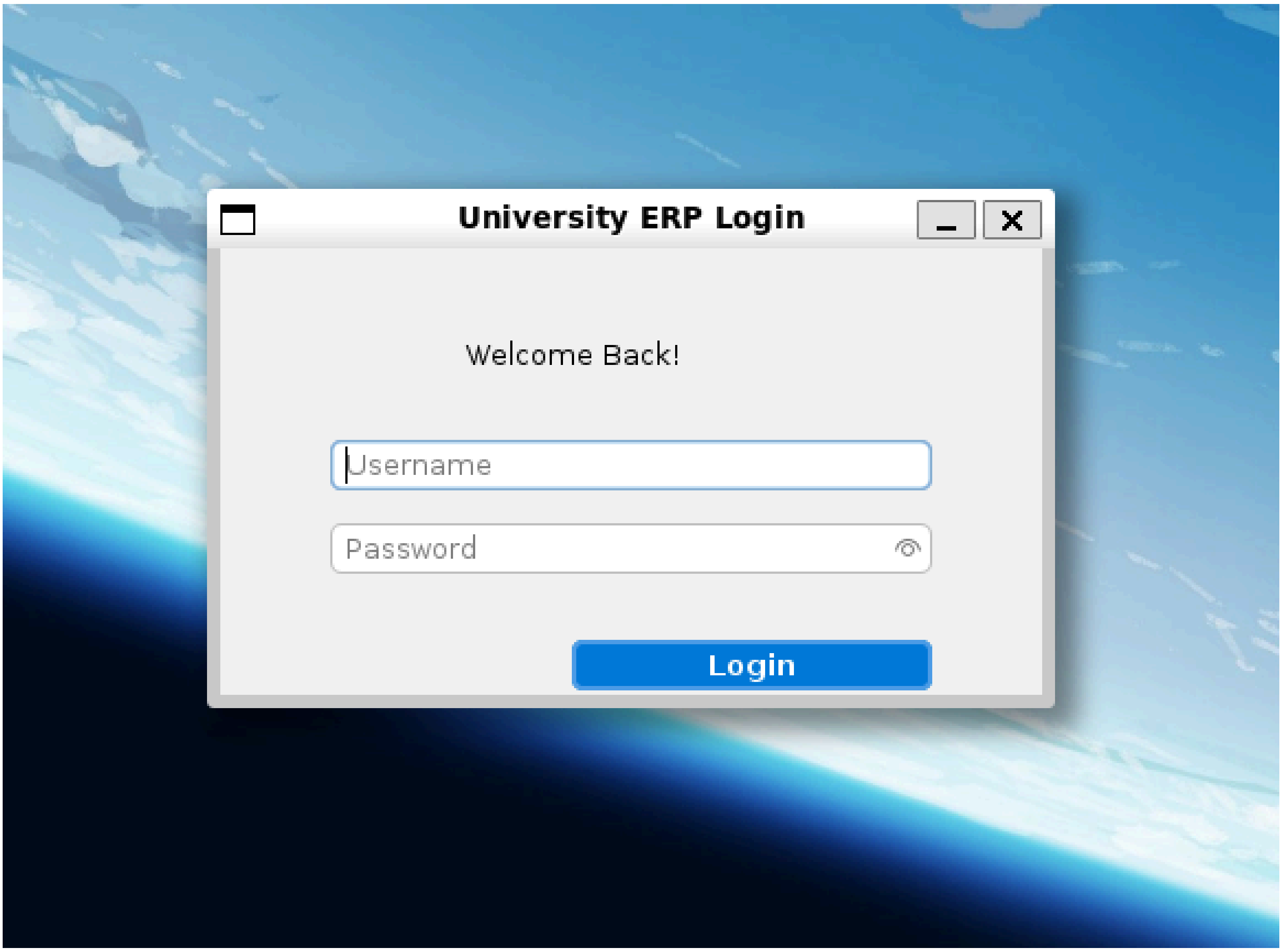


Fig: Login Screen

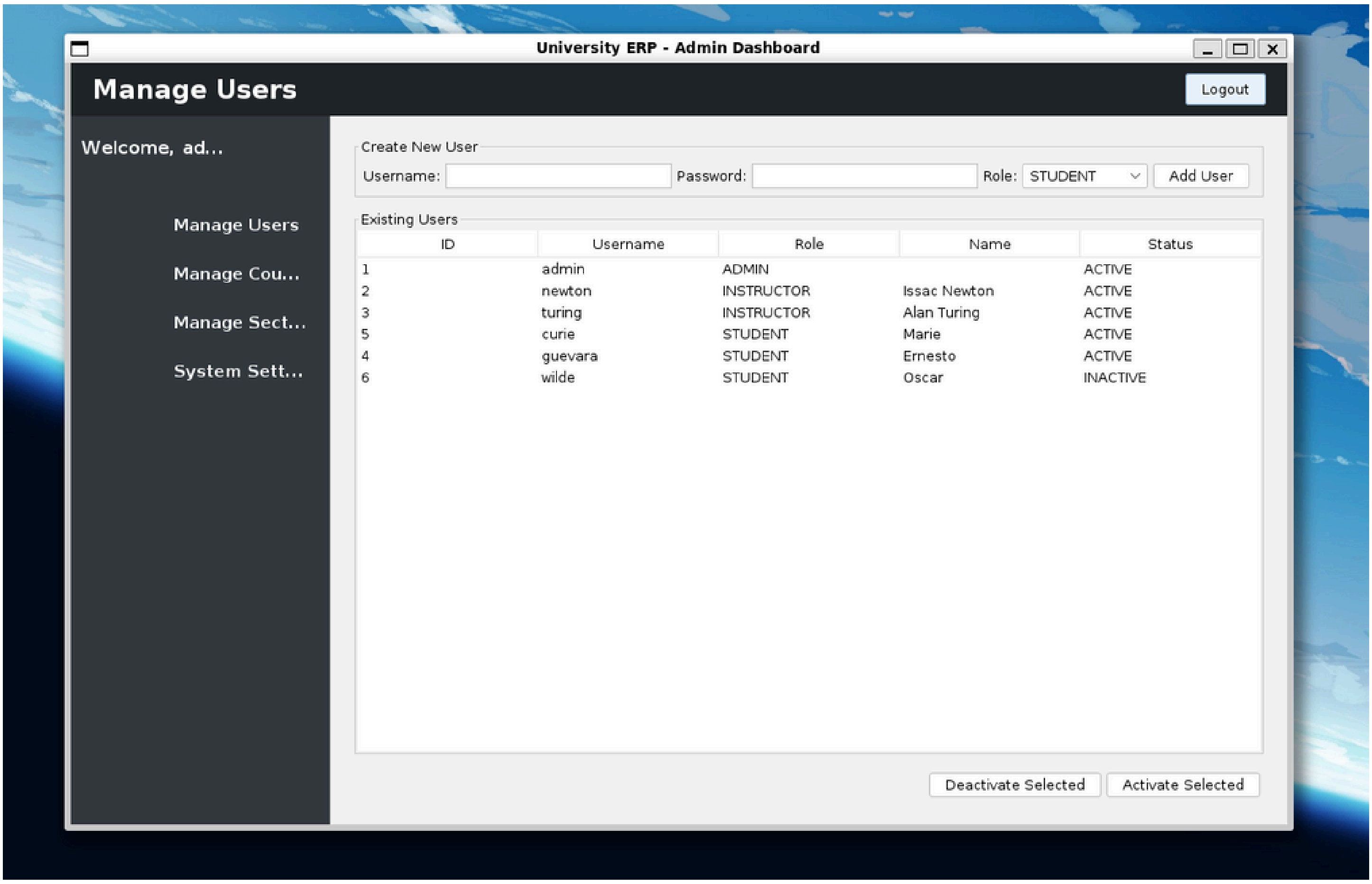


Fig: Admin Dashboard - Manage Users Tab

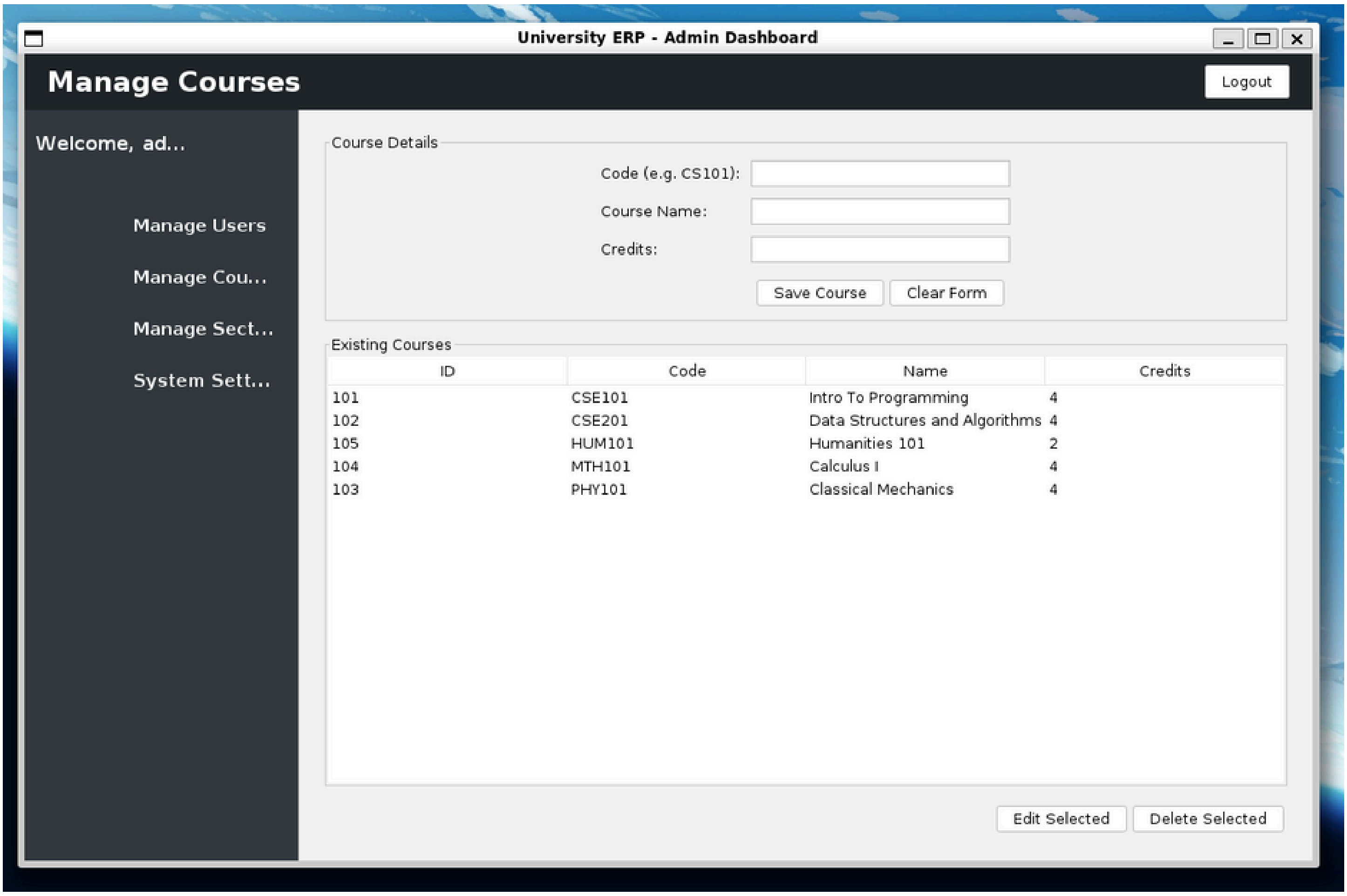


Fig: Admin Dashboard - Manage Courses Tab

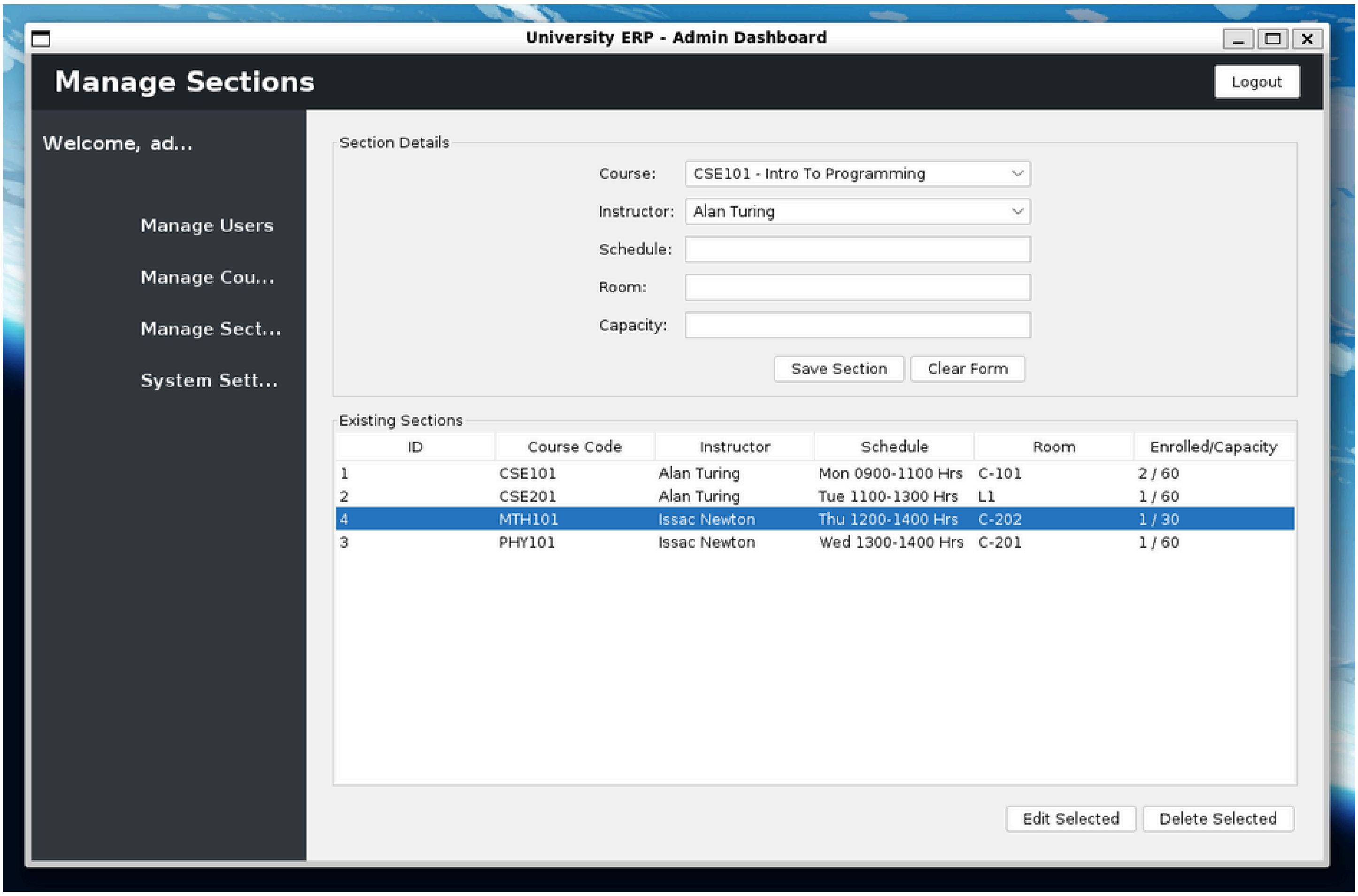


Fig: Admin Dashboard - Manage Sections Tab

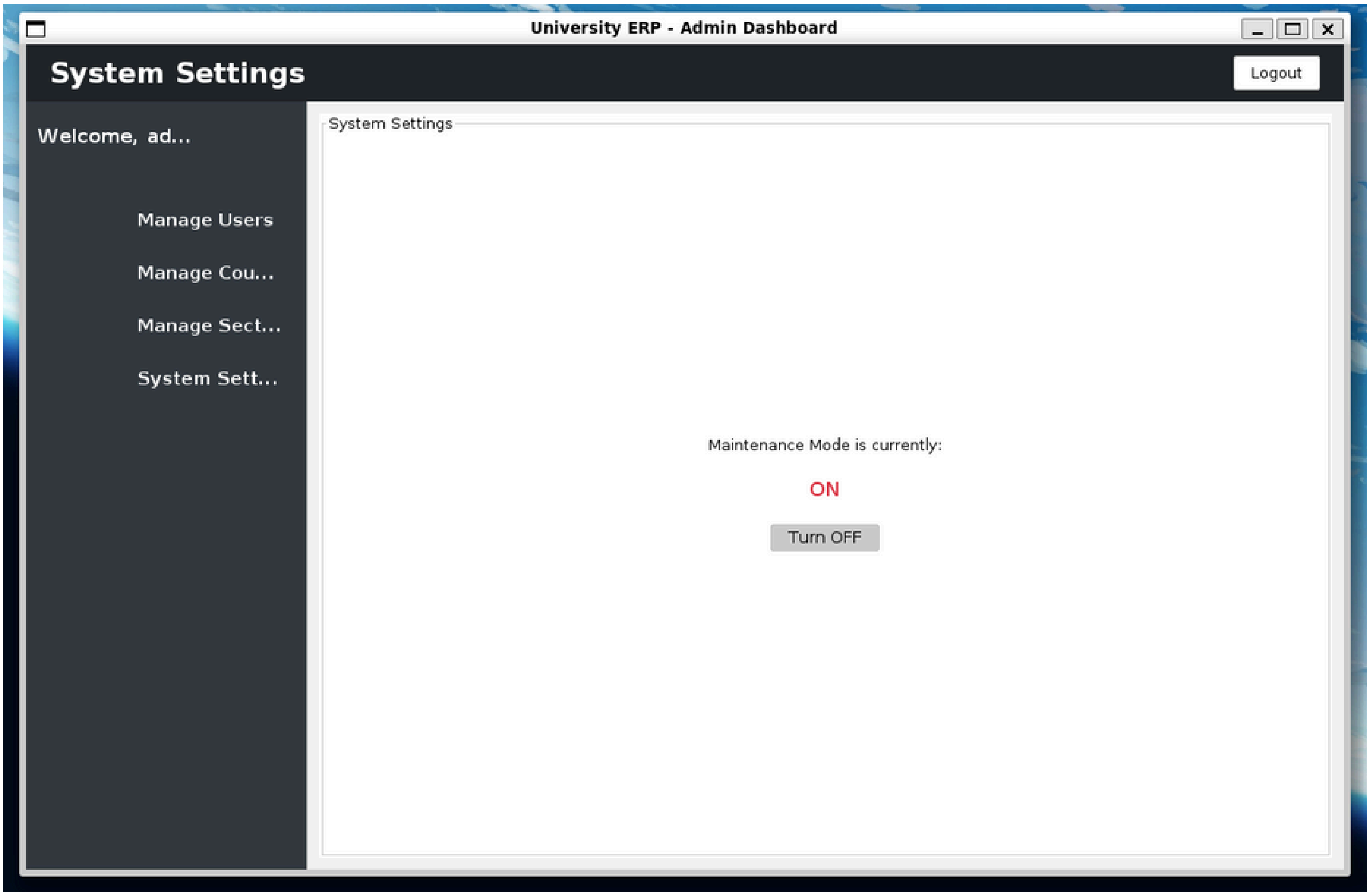


Fig: Admin Dashboard - Maintenance Mode Settings

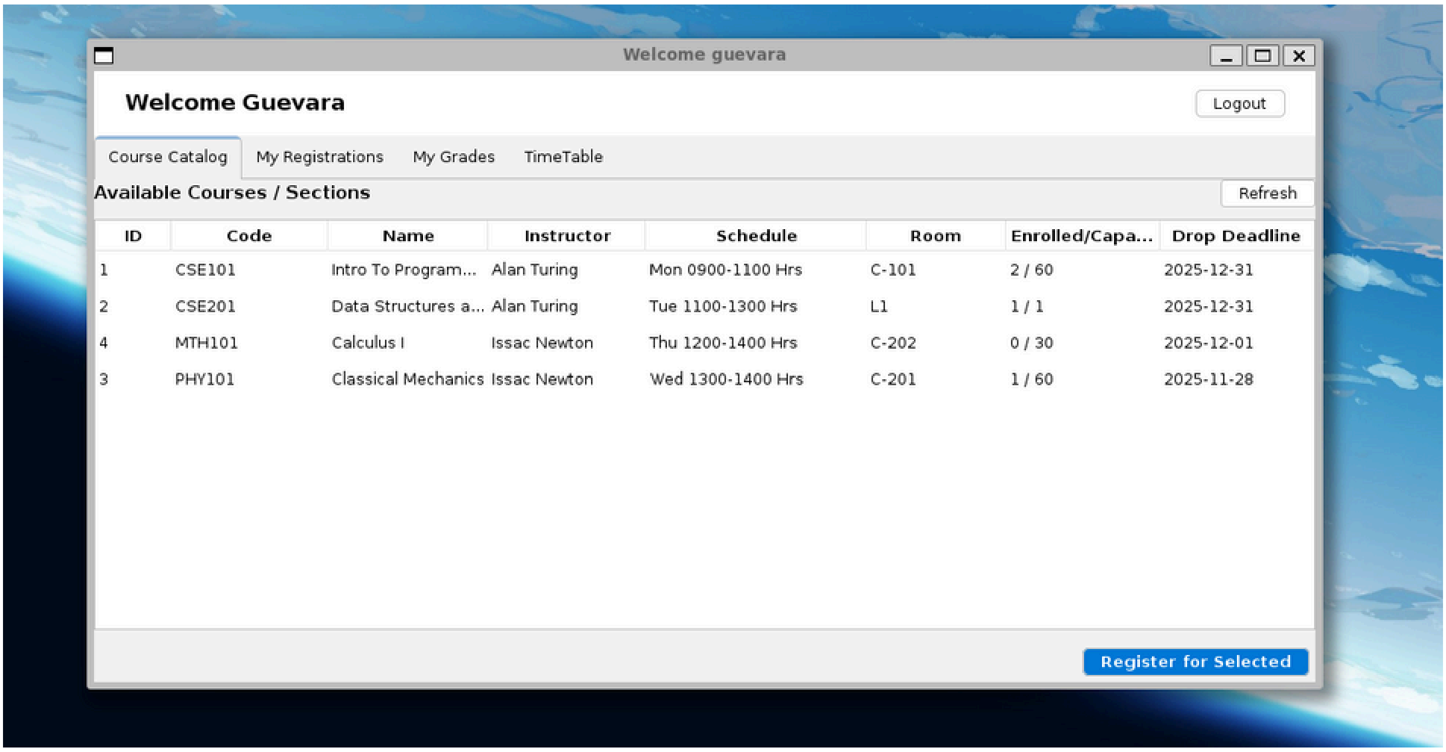
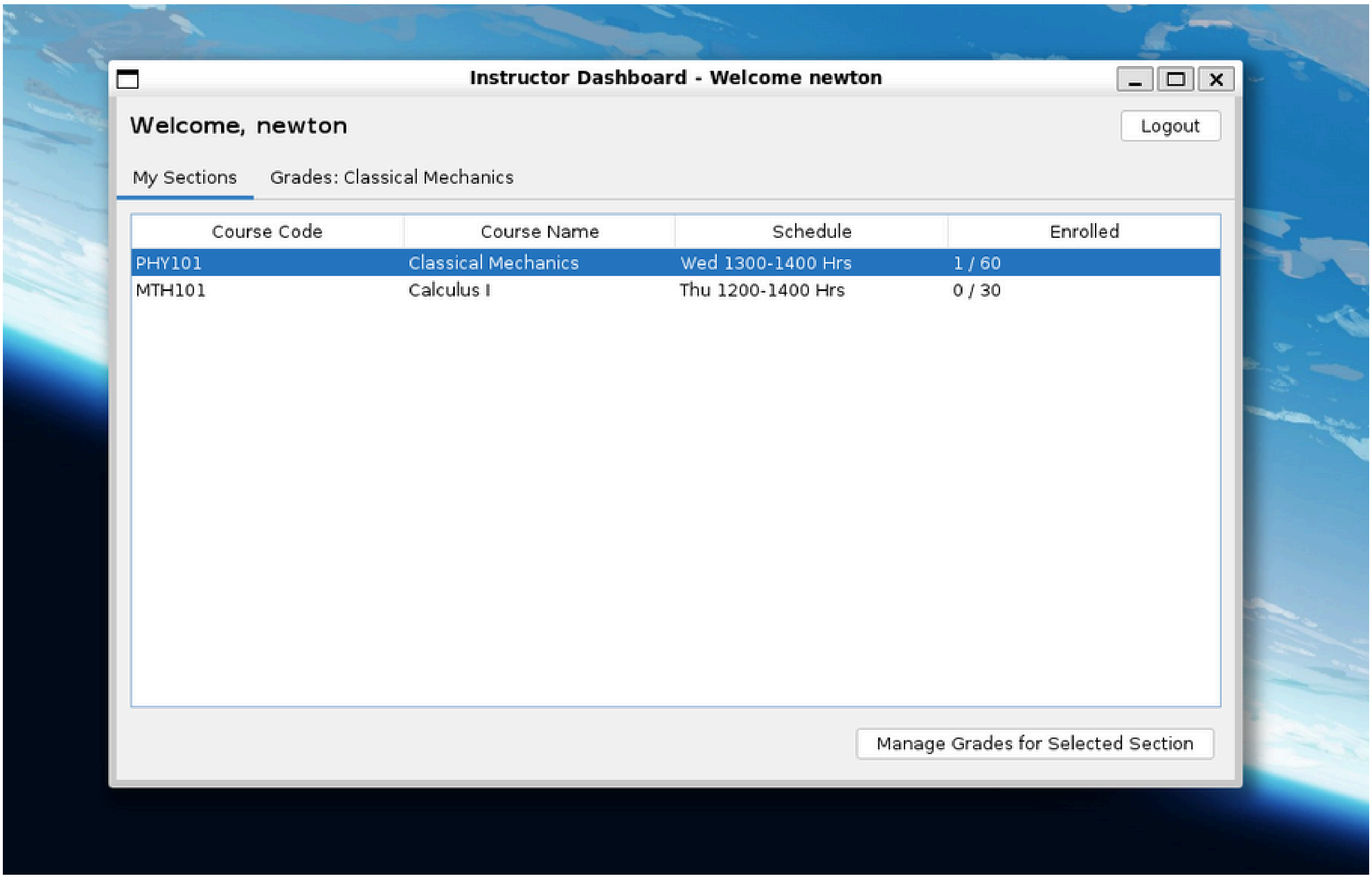


Fig: Student Dashboard



Instructor Dashboard